Edward O'Dwyer

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Technical Skills

Languages: Java, Python, C/C++, SQL (PostgreSQL & SQLite), Shell script, Haskell
Dev. Tools/Frameworks: Git/GitHub, Linux/UNIX, Neovim, Valgrind, ASan/UBSan, Maven, JUnit, NumPy, Pandas, REST
Concepts: Low-Latency Systems, Concurrency, API Design, Parallel Computing

Education

University of Warwick — Coventry, UK
Bachelor of Science (Honours) in Computer Science — *Sep 2024 – Jul 2027*
- Expected: First | Modules: Advanced Comp. Architecture, Compilers, High Performance Computing (4th-year module)
- Dissertation: *eDSL for Compile-Time Task Dependency Graph Optimisation* (C++)

The American School in London — London, UK
High School Diploma — *Aug 2017 – Jun 2024*

Experience

Data Engineer Intern — Jul 2025 – Sep 2025
Homey Technology | *Python, Pandas, NumPy, Plotly, Streamlit, PostHog, PostgreSQL* — *London, UK*
- Drove a **97% reduction** in KPI dashboard latency via a **concurrent caching system** with per-query mutex locks
- Built a company-wide KPI dashboard unifying **5 data sources**, architected so new sources plug in easily
- Applied **correlation analysis** and lines of best fit to KPI trends, producing **20+ strategic forecasts**

Projects

Airport Traffic Simulation | *Java, JavaFX, Maven, JUnit* — *Departmental Prize* | Jan 2026 – Mar 2026
- Architected the **MVVM** design for a **team of 6**, using immutable state snapshots to decouple engine and UI
- Built the simulation engine's **lock-free** threading model using AtomicReference and a concurrent command queue
- Implemented a **headless mode** that skips snapshot allocation and rendering, running 720 simulated hours in **0.3s**

Crowd Dynamics and Optimisation Engine | *Python, Numba, NumPy* — *Built in 24h* | Feb 2026
- Engineered a custom **Verlet integration physics engine**, utilising **JIT compilation** to simulate 2,000+ agents at 60 FPS
- Implemented **Spatial Partitioning** via Cell Lists to optimise collision detection from $O(N^2)$ to $O(N)$ ensuring performance
- Developed a **Monte Carlo Search** algorithm to optimise barrier placement, achieving a ∼60% reduction in crowd pressure

Hybrid Chess Engine Framework | *C++, Python, Numba, SFML* — Dec 2025 – Jan 2026
- Designed a hybrid engine letting users inject **Numba-compiled machine code** into a low-latency C++ backend
- Evolved search from Minimax to **Negamax with Alpha-Beta**, adding **Quiescence Search** to mitigate the horizon effect
- Built move generation and GUI from scratch in C++/SFML, using **bitboards** for fast validation within 150MB of memory

Network Attacks Identifier | *C, POSIX Threads* — Nov 2025 – Dec 2025
- Architected a **single-producer, multi-consumer** pipeline with a **worker pool sized to core count** for true parallelism
- Engineered a lock-synced **ring buffer** for passing packets, eliminating per-packet allocation and exploiting cache locality
- Applied **fine-grained locking** over shared detection state, verifying **memory safety** under attack simulation with Valgrind

A=B Esolang Interpreter | *Haskell, Cabal, Stack* — Mar 2025 – Apr 2025
- Built an interpreter for the A=B (a to b) esoteric language using Haskell, featuring a **custom parser**
- Achieved correct interpretation of **100%** of official A=B examples, demonstrating deep understanding of **idiomatic** Haskell

Awards

UKIEPC 2026 – 2nd of 22 Warwick Teams — *Qualified for NWERC* | Oct 2026
- Competed in the UK & Ireland Programming Contest, an ICPC-format algorithmic competition for university teams

IC Algothon 24 – Finalist | *Python, NumPy, Pandas, Matplotlib, JavaScript* — Nov 2024
- Placed 9th out of ∼ 200 teams (**top 4.5%**) in Europe's largest algorithmic trading hackathon (**25% admittance rate**)

Leadership

Warwick Hackathon Organiser — Oct 2024 – Present
Warwick Hackathon (WHACK)
- Co-lead organiser of **WHACK 2026**: **nearly tripled sponsorship to £26k** and **doubled capacity to 400** year-on-year
- Owned sponsor negotiations end-to-end, restructuring tiers around a **£10k title package** to anchor gold pricing
- Built a **Binomial turnout model** with a CLT approximation to size overbooking: **521 places** at **0.1%** overflow risk

Academic Executive Officer — Oct 2024 – Jun 2026
University of Warwick Computing Society
- Supported **dozens of students** through recruitment with resume reviews, mock interviews, resources and workshops
- Partnered with **10+ societies** on joint events, reaching compromises through responsive, needs-focused collaboration
- Led a Chess Engine Competition for **40 competitors**, securing Optiver as sponsor and building the framework it ran on